Lesson 5.11 - Bayes' Theorem of Probability with Tree Diagrams

> **Video:** Bayes' Theorem of Probability With Tree Diagrams & Venn Diagrams by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Deriving Bayes' theorem

Start from two conditional probabilities:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Both give the same joint probability:

$$P(A \mid B)\,P(B) = P(A \cap B) = P(B \mid A)\,P(A)$$

Divide by $P(B)$:

$$\boxed{P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}}$$

Bayes' theorem gives a conditional probability when you know the **reverse** one, $P(B \mid A)$.

Example 1 - slips of paper (checking the formula)

A bottle holds slips numbered 1-9, each equally likely to be drawn. $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7, 8, 9\}$, $A \cap B = \{4, 5\}$

Venn regions: A only = {1, 2, 3}; both = {4, 5}; B only = {6, 7, 8, 9}.

- $P(A) = \frac{5}{9}$, $P(B) = \frac{6}{9}$
- $P(B \mid A)$: of A's 5 numbers, 2 are in B, so $\frac{2}{5}$

Using Bayes:

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)} = \frac{\frac{2}{5} \cdot \frac{5}{9}}{\frac{6}{9}} = \frac{2/9}{6/9} = \frac{2}{6} = \frac{1}{3}$$

Check with the direct formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{2/9}{6/9} = \frac{1}{3} \checkmark$$

Example 2 - medical screening

- 12% of men develop prostate cancer: $P(C) = 0.12$
- With cancer, the test is positive 95% of the time: $P(+ \mid C) = 0.95$
- Without cancer, there is a 6% false-positive rate: $P(+ \mid C') = 0.06$

Find $P(C \mid +)$, the chance of having cancer given a positive test.

Step 1 - tree diagram

```
                0.95  positive   →  0.12 × 0.95 = 0.1140
        C  <
 0.12 /         0.05  negative   →  0.12 × 0.05 = 0.0060
start
 0.88 \         0.06  positive   →  0.88 × 0.06 = 0.0528
        C' <
                0.94  negative   →  0.88 × 0.94 = 0.8272
```

Step 2 - total probability of a positive test

A positive result happens through either branch, so add them:

$$P(+) = P(C)P(+ \mid C) + P(C')P(+ \mid C') = 0.1140 + 0.0528 = 0.1668$$

Step 3 - Bayes

$$P(C \mid +) = \frac{P(+ \mid C)\,P(C)}{P(+)} = \frac{0.95 \times 0.12}{0.1668} = \frac{0.114}{0.1668} \approx 0.683$$

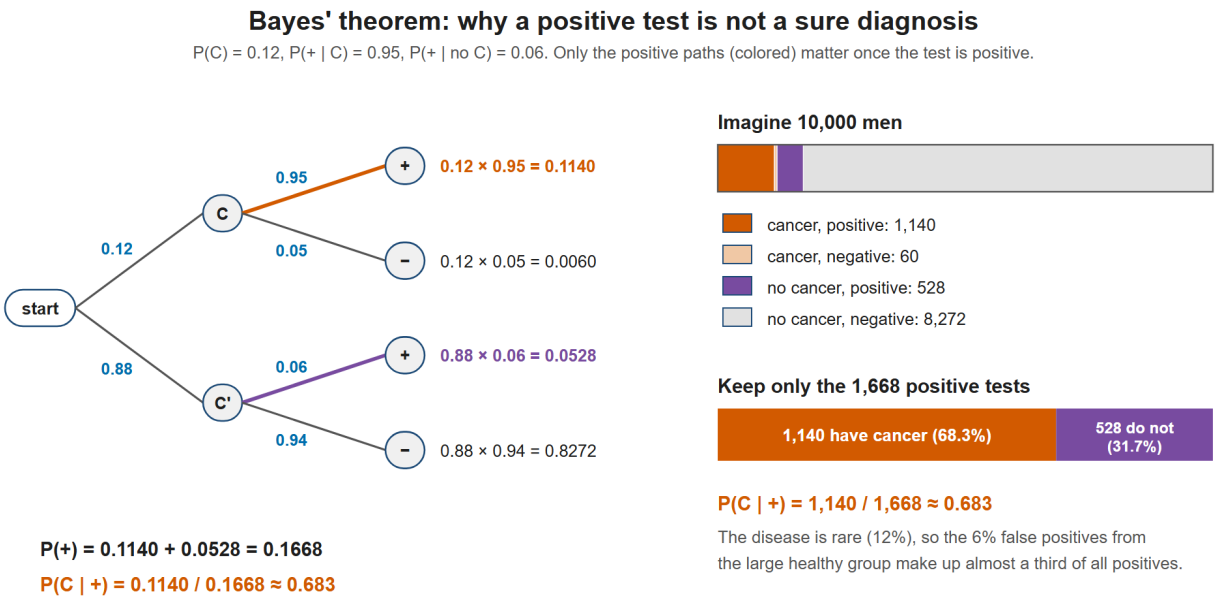
A man with a positive test has about a **68.3%** chance of having cancer, so about **31.7%** of positive results are false alarms ($1 - 0.683$).

Same answer with counts (imagine 10,000 men)

	Positive	Negative	Total
Cancer (12%)	1,140	60	1,200
No cancer (88%)	528	8,272	8,800
Total	**1,668**	8,332	10,000

$$P(C \mid +) = \frac{\text{cancer and positive}}{\text{all positive}} = \frac{1140}{1140 + 528} = \frac{1140}{1668} \approx 0.683 \checkmark$$

The choice of 10,000 doesn't matter, because only the ratio counts.



Out of 10,000 men, 1,668 test positive, but only 1,140 of them have cancer. Restricting to the positive tests gives P(C | +) = 1,140 / 1,668 ≈ 0.683.

Key takeaways

- $P(A \mid B) = \dfrac{P(B \mid A)P(A)}{P(B)}$, which turns a known conditional probability around.
- The denominator often has to be built from a tree: $P(B) = P(B \mid A)P(A) + P(B \mid A')P(A')$.
- A positive test does **not** guarantee the condition. Rare conditions plus false positives can produce many false alarms.
- Converting to counts (e.g. 10,000 people) is an easy way to check the answer.